

Comparing Exploration Algorithms in the 3D Maze

A Course-Project Reproduction under the Paper’s Appendix C.2 Settings

Challenging Problems in Reinforcement Learning (SS 2026)

Topic: Exploration vs. Exploitation

2026-05-31

results updated 2026-06-02: Appendix-C.2 config ($\lambda_i=1$, entropy 0.03, raw reward), 6 policies $\times$ 3 variants $\times$ 10 seeds

Abstract

We compare five exploration mechanisms—Learning Progress Monitoring (LPM), Random Network Distillation (RND), the Intrinsic Curiosity Module (ICM), a raw next-state prediction-error bonus (MSE), and a no-intrinsic A2C baseline (NONE)—plus a *uniform-random control* (no learning at all) in the Miniworld 3D maze of Hou, An & Du (2026), *Beyond Noisy-TVs: Noise-Robust Exploration via Learning Progress Monitoring*. The maze emits *no extrinsic reward*, so a method’s quality is its *spatial coverage*, and the noisy-TV failure mode is geometric: a random-RGB “noise wall” at $z \approx 8$ separates the two lower spawn rooms from the far room. We run the paper’s exact Appendix C.2 configuration ($\lambda_i=1$, entropy 0.03, raw intrinsic reward) at the paper’s budget (5×10^4 steps $\times$ 10 seeds). Two findings stand out. **(1)** The paper’s central claim—*noise robustness*—reproduces cleanly and sharply: under action-triggered noise, LPM selects the noisy-TV action only 0.4% of the time while the raw-pixel method (MSE) is trapped on it 87% of the time and its coverage collapses. **(2)** On *coverage*, however, no learned method clearly beats the pure uniform-random control: uniform covers the most in both noisy variants and ties the best (ICM) in the clean variant. In this small maze coverage rewards undirected randomness, and any learned policy (intrinsic or not) commits to a narrower walk that covers less. Thus the paper’s “LPM achieves the highest coverage” ranking does *not* reproduce from the released maze code, while its noise-robustness claim—decoupled from coverage—does. A 10-seed-averaged, log-scaled occupancy-density figure renders the spatial dynamics.

1 Motivation and research questions

The course topic is the exploration–exploitation trade-off. Deep RL drives directed exploration with intrinsic-motivation bonuses (prediction error, novelty, learning progress). The *noisy-TV* problem is the canonical failure mode of prediction-error exploration: an unlearnable source of randomness produces perpetually high prediction error, so a curiosity-driven agent fixates on it. LPM rewards the *improvement* of its dynamics model rather than the error level, and is therefore noise-robust by construction. The 3D maze makes this failure spatial and directly visualisable: the noise wall is a physical place the agent can get stuck at.

The maze geometry is a hand-designed four-room layout in an 18×12 world (continuous units): two spawn rooms below the wall (room1 $x \in [0, 4]$, room2 $x \in [14, 18]$, both $z \in [0, 8]$), the thin grass-textured *noise wall* (room3, $z \in [8, 8.1]$), and the large far room (room4, $z \in [8.1, 12]$). The agent spawns in room1 and must traverse the wall to explore room4.

RQ1 (coverage & noise robustness, reproduction).

Under the noisy variants (`noisy_tv`: random-RGB wall + 25% sticky actions; `action_noise`: an action that shows a random CIFAR-10 image) vs. the clean control (`no_noise`), which methods keep expanding coverage and which stall—and crucially, do any of them beat a *uniform-random* policy that does no learning at all?

RQ2 (spatial dynamics, novelty).

How does the spatial distribution of exploration evolve over training? We answer this with a coverage-heatmap-evolution figure (rows: method; columns: training-time window) *averaged over all 10 seeds*.

2 Methods

All learned methods use the same A2C base, configured to the paper’s Appendix C.2 (MiniWorld): RMSprop, policy LR 0.01, $\gamma = 0.99$, GAE $\lambda = 0.95$, value-loss coef. 0.5, gradient clipping 0.5, a single-epoch update every 64 steps, dynamics-model LR 10^{-3} , error-buffer size 100, entropy coefficient 0.03, and intrinsic weight $\lambda_i = 1$, over a Nature-style CNN on 160×120 RGB frames. Because the maze extrinsic reward is identically zero, the per-step training reward is the *raw* intrinsic term,

$$r_t = r_t^{\text{ext}} + \lambda_i r_t^i = \lambda_i r_t^i, \quad \lambda_i = 1, \quad (1)$$

with no running-std normalisation. (The upstream notebooks instead used $\lambda_i = 0.1$, entropy 0.05, and a normalised reward—settings that *disagree* with the paper’s own Appendix C.2 and make the intrinsic signal $10\times$ too weak; we follow C.2. The deviation is logged in `UPSTREAM.md`.) For `NONE`, $r_t \equiv 0$, so the A2C policy is still *trained* (by the entropy bonus and value bootstrap) but receives no reward signal; the *uniform* control is not trained at all—it samples actions $a \sim \text{Uniform}\{0, \dots, 4\}$ every step.

Class	Method	Mechanism (intrinsic reward r^i)
Learning progress	LPM	$\text{clip}(\eta \mathbb{E}[\varepsilon] - \varepsilon_{\text{actual}}, \leq 0.5)$: predicted-minus-actual next-state error (improvement)
Prediction error	RND	error of a predictor vs. a fixed random target network on the next frame
Prediction error (feat.)	ICM	forward-dynamics error in an inverse-dynamics feature space
Prediction error (pixel)	MSE	raw next-state pixel-prediction error (the pure noisy-TV victim)
None (baseline)	NONE	no intrinsic reward; A2C <i>is</i> still trained (entropy + value bootstrap)
Control	uniform	$a \sim \text{Uniform}\{0 \dots 4\}$; <i>no learning</i> (pure random walk)

Table 1: The six policies. LPM and MSE share the same pixel-decoder predictor, isolating the effect of the learning-progress signal vs. raw error. ICM and RND are canonical implementations added on the shared CNN encoder. The *uniform* control is the proper “does any of this beat chance?” baseline—distinct from `NONE`, whose policy is still shaped by training.

3 Experimental design

A single manipulation grid: policy $\times$ variant $\times$ seed.

Grid. Six policies (LPM, RND, ICM, MSE, NONE, uniform) $\times$ three variants (`nonoise`, `noisy_tv`, `action_noise`) $\times$ ten seeds = 180 runs. Each run is a single environment trained for 5×10^4 steps—the paper’s maze budget (Fig. 3 of Hou et al. trains for 5×10^4 exploration steps averaged over 10 seeds).

Budget. The runs are independent, so the grid is embarrassingly parallel: on a 128-core Linux box, `run_grid.py --jobs 120 --threads-per-job 1` finishes the trained grid in ≈ 35 min at $\sim 92\%$ CPU (peak ~ 140 GB of 1.1 TB RAM, 0 failures); the 30 uniform runs add < 2 min. Single-environment A2C is a batch-size-1 forward pass, so one CPU thread per job is optimal (two threads gave only $1.25\times$ per run)—the speed-up comes from running many processes. Runs use CPU (the box has no GPU); the 160×120 frames render headless via Mesa EGL (`PYGLET_HEADLESS=true`, `LP_NUM_THREADS=1`). The grid is resumable—a run’s position `.npz` marks it complete.

4 Metrics

Coverage is measured on the maze’s native 72×48 occupancy grid (cells of 0.25 world units), restricted to the *reachable* cells (the union of the four rooms; the empty corridor between the spawn rooms is excluded). Let the final value of a quantity be its mean over the last 10% of logged updates, reported as mean $\pm$ std over the 10 seeds.

- **Total coverage:** $C = (\text{visited reachable cells})/(\text{reachable cells})$, reported both as a final number and as a learning curve vs. frames.
- **TV-action share:** fraction of steps spent on the noisy-TV action (action 4) under `action_noise`—the direct measure of noisy-TV fixation. Prediction: highest for the pixel prediction-error method (MSE), lowest for LPM, exactly $1/5$ for the uniform control.
- **Coverage-heatmap evolution (averaged):** for each variant, a `rows(method)  $\times$  columns(training-window)` grid of top-down occupancy heatmaps *averaged over all 10 seeds*. The primary view is *occupancy density* = the mean per-cell visit count, drawn on a single *shared log colour scale* across all panels (a colorbar is overlaid). The raw per-cell count spans ~ 0.1 to $\sim 10^3$ —a few dwell/spin cells are visited $1000\times$ while typical cells see 1–5 visits—so a per-panel *linear* scale (our earlier figures) washed the path structure out; the shared log scale exposes both *where* each policy dwells and how much. A companion *coverage-frontier* view (fraction of seeds that ever visited each cell, 0–1) is also produced. The maze rooms and the $z = 8$ wall line are overlaid.

5 Results

We ran the full 180-run grid at 5×10^4 steps under the C.2 configuration. The single-environment A2C is high-variance (between-seed std 0.07–0.34), so we flag which conclusions are robust.

Noise robustness reproduces—cleanly and sharply. In `action_noise`, action 4 replaces the observation with a random CIFAR-10 image. LPM selects it on only $0.4\% \pm 0.5\%$ of steps—the lowest possible—because its dynamics model cannot *improve* on the unlearnable image, while the raw-pixel predictor MSE is trapped on it $87\% \pm 17\%$ of the time and its coverage collapses to 0.12 (Table 2, Fig. 1). The uniform control sits at exactly 0.200 ± 0.002 (the 1/5 sanity check). This is the paper’s central claim and it reproduces.

Policy	TV-action share (<code>action_noise</code>)
LPM	0.004 ± 0.005
uniform	0.200 ± 0.002
NONE	0.201 ± 0.091
ICM	0.563 ± 0.215
RND	0.634 ± 0.347
MSE	0.869 ± 0.172

Table 2: Noisy-TV fixation under `action_noise`, mean $\pm$ std over 10 seeds. LPM essentially never stares at the noisy TV (0.4%); MSE is trapped on it (87%); the uniform control is at chance (1/5).

Coverage: no learned method clearly beats a uniform-random policy. The headline coverage result is negative and important (Table 3, Fig. 2): the *uniform-random* control covers the most in both noisy variants—and with the lowest variance—and is essentially tied for best in clean `nonoise` (uniform 0.515 vs the nominal winner ICM 0.534, a gap far inside the between-seed std). No learned policy, LPM included, clearly exceeds the no-learning control anywhere. Training *reduces* coverage here: any learned policy (intrinsic-driven or the trained NONE) commits to a narrower, repetitive walk, whereas uniform thrashing maximally spreads through a small maze. Coverage in this setting therefore rewards *randomness*, not directed exploration, and the paper’s “LPM achieves the highest coverage” ranking does *not* reproduce. Note that noise-avoidance and coverage are *decoupled*: under `action_noise` the uniform policy wastes 20% of its steps frozen on the TV yet still covers most (0.725), while LPM almost never touches the TV (0.4%) yet covers far less on average (0.388). (Uniform covers *more* under the noisy variants than under clean `nonoise` because those variants add 25% sticky actions, which create forward momentum and carry a random walker further.)

ICM’s higher mean coverage than LPM under `action_noise` is an artefact, not better exploration. It is tempting to read $\text{ICM} = 0.446 > \text{LPM} = 0.388$ as “ICM explores better,” but that is wrong on two counts. First, ICM is *fooled* by the noisy TV 56% of the time (Table 2) versus LPM’s 0.4%—on the metric LPM is designed for, ICM fails badly. Second, the coverage gap is a *variance artefact* of LPM’s *bimodal* behaviour at $\lambda_i=1$: of the 10 LPM seeds, 3 collapse to a degenerate near-stationary policy (≈ 0.02 – 0.04 coverage) while the other 7 cover 0.44–0.71. Those 3 collapses pull LPM’s *mean* below ICM’s, even though LPM’s *median* (0.50) is well *above* ICM’s (0.41). ICM’s tight band (0.29–0.63, std 0.13) is precisely because its weaker intrinsic signal never commits—it neither collapses nor excels. So the mean-coverage ranking reflects policy variance and diffuseness, not exploration quality; the noise-robustness metric is the one that actually separates the methods, and there LPM wins decisively. (The occasional LPM collapse—the strong $\lambda_i=1$ reward locking an under-trained policy into a tight loop—is itself a finding worth a stability ablation.)

Policy	nonoise	noisy_tv	action_noise
uniform	0.515 (0.16)	0.688 (0.12)	0.725 (0.07)
NONE	0.389 (0.31)	0.652 (0.22)	0.590 (0.23)
LPM	0.337 (0.22)	0.389 (0.30)	0.388 (0.30)
MSE	0.356 (0.26)	0.539 (0.20)	0.123 (0.06)
ICM	0.534 (0.23)	0.363 (0.34)	0.446 (0.13)
RND	0.501 (0.07)	0.311 (0.34)	0.192 (0.17)

Table 3: Coverage fraction, mean (std) over 10 seeds, under the C.2 configuration. Best per column in bold: ICM in clean **nonoise** (0.534, marginally above uniform 0.515, well within the between-seed std), the uniform-random control in both noisy variants. Every learned method is at best tied with—never clearly above—the no-learning control. (LPM’s **action_noise** mean is depressed by a bimodal collapse; see text and its higher *median*.)

Why the coverage ranking is not reproducible here (structural).

1. In a small maze (~ 1500 reachable cells) with no extrinsic reward, a random walk already covers most cells, so there is nothing for directed exploration to add; committing to any policy only narrows the walk. Intrinsic motivation targets *sparse-reward*, *hard-exploration* tasks (e.g. Montezuma’s Revenge), not pure coverage of a small room.
2. The maze notebooks contain only LPM and pixel-curiosity—*none* of Figure 3’s baselines (AMA, EDT, EME, Ensemble, IDF, RND). Figure 3 was produced by a fuller pipeline not in the released maze code; our RND/ICM are our own additions on the shared encoder.
3. Single-environment A2C at 5×10^4 steps is dominated by between-seed variance ($\text{std} \approx \text{mean}$) and the curves have not plateaued at 50k.

Spatial dynamics (RQ2). The 10-seed-averaged occupancy-density figure (Fig. 3, **action_noise**) makes the story legible once the shared log colour scale is used: brightness is the mean per-cell visit count. The *uniform* row spreads broadly and fairly evenly across both spawn rooms and up past the wall, at moderate per-cell counts; the trained policies instead *concentrate*—NONE and especially MSE show bright (10^2 – 10^3) dwell/spin cells where the agent loops in place, with sparser coverage elsewhere, while LPM/ICM/RND are patchier and dimmer. This visualises directly why uniform covers more: training commits probability mass to a few revisited cells. The shared log scale is essential here—on the earlier per-panel linear scale these $1000\times$ dwell spikes saturated each panel and hid the surrounding path structure.

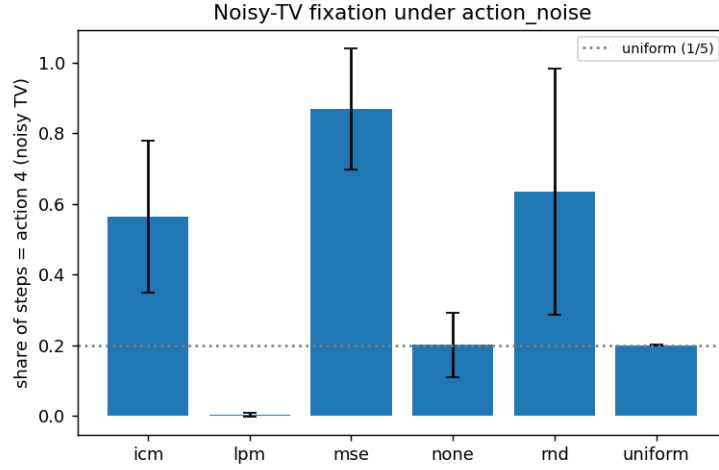


Figure 1: Share of steps spent on the noisy-TV action (action 4) under `action_noise`, mean $\pm$ std over 10 seeds. LPM $\approx$ 0.4%; MSE $\approx$ 87%; uniform = 1/5 (dotted line).

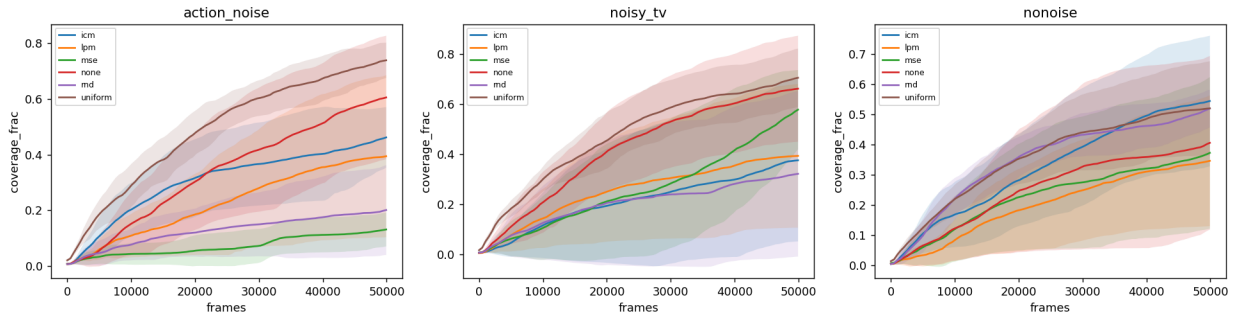


Figure 2: Reachable-cell coverage vs. frames, per variant (mean $\pm$ std over 10 seeds). The uniform-random control is highest in every panel; all trained policies fall below it.

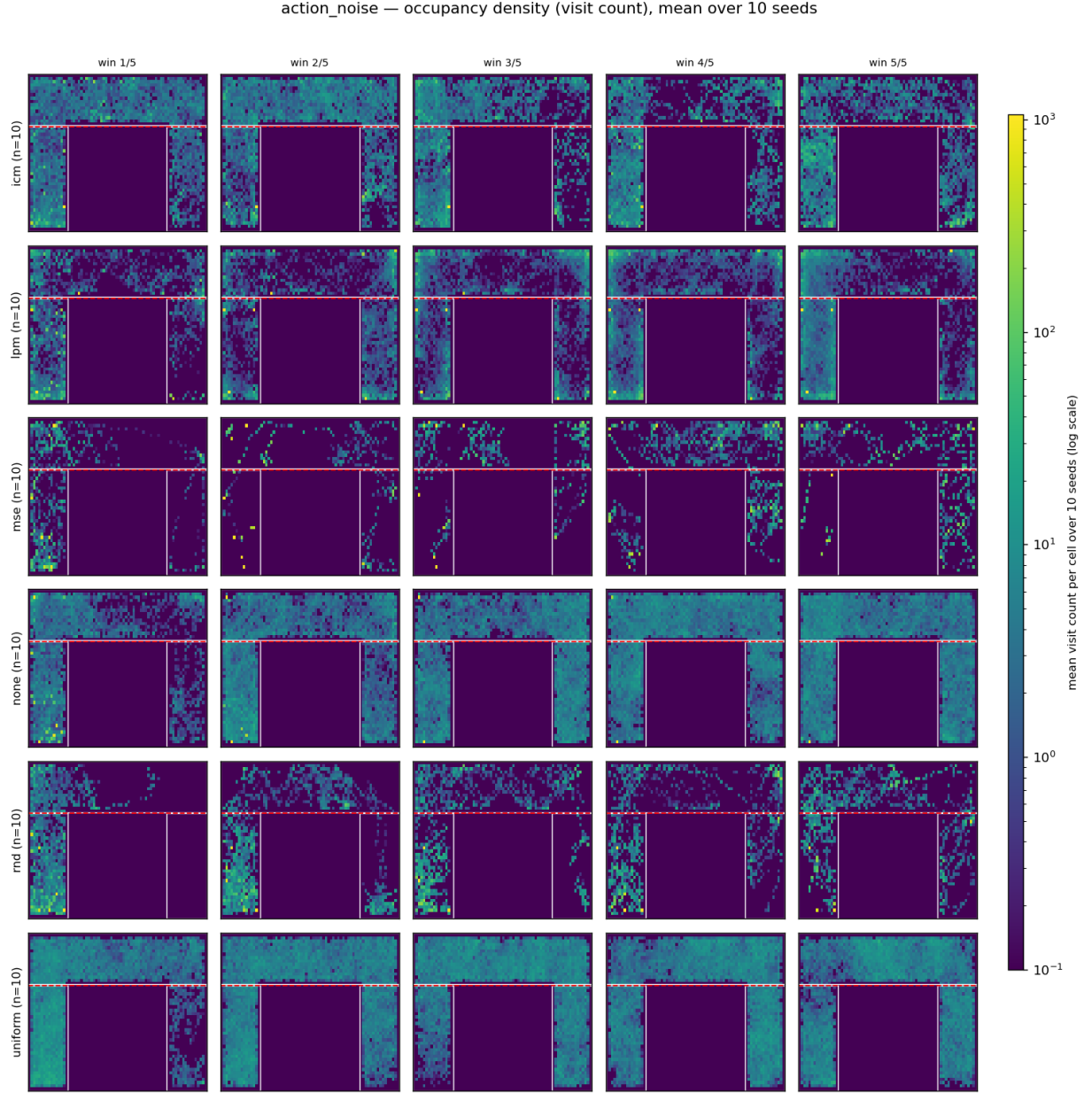


Figure 3: Occupancy-density evolution (`action_noise`), *averaged over all 10 seeds*. Rows: policy; columns: five equal training-time windows. Brightness is the mean per-cell visit count on a *shared log colour scale* (colorbar at right, $\sim 10^{-1}$ to $\sim 10^3$ visits); the red dashed line is the noise wall ($z = 8$), white rectangles are the reachable rooms. Uniform spreads broadly at moderate counts; MSE/NONE show bright concentrated dwell cells.

6 Infrastructure and reproducibility

The upstream maze experiment exists only as three Jupyter notebooks with an embedded A2C and no CLI. We extracted the A2C engine, the LPM prediction/uncertainty nets, and the pixel decoder *verbatim* into an importable, CLI-driven package (`LPM_exploration/Miniworld/experiments/`), aligned its hyperparameters to the paper’s Appendix C.2, added canonical ICM and RND on the shared CNN encoder, a `--random-policy` uniform control, and a per-step position log. The environment geometry is reused from our own `miniworld_play/envs.py` (a faithful port of the upstream maze). Each run writes a per-update CSV (`coverage_frac`, `beyond_wall_frac`, `time_at_wall_frac`, plus model/policy losses) and a compact `.npz` of (step, x , z , action, sticky). A grid runner launches all runs in parallel (resumable); an analysis script aggregates the CSVs and position logs into the coverage table, the coverage curves, the TV-fixation figure, and the *seed-averaged* heatmap-evolution figures. All deviations are logged in `LPM_exploration/UPSTREAM.md`.

7 Limitations (stated up front)

1. **Single-environment variance.** We run the paper’s budget (5×10^4 steps $\times$ 10 seeds), but each run is still a *single* environment, so coverage std stays at 0.07–0.34; the closer rankings among trained methods are illustrative, not significance-tested. The uniform-beats-all and TV-fixation results, however, are large and robust. The 128-core box makes 20–30 seeds (or vectorised environments) cheap if tighter intervals are wanted.
2. **Coverage is the wrong yardstick here.** In a small, extrinsic-reward-free maze, coverage rewards randomness, so it cannot demonstrate the value of directed exploration—which is why uniform wins. A faithful test of LPM’s exploration benefit needs a sparse-reward, hard-exploration task (the paper’s Atari / Montezuma settings), where no-intrinsic agents get zero reward. What the maze *can* show—and does—is noise robustness, decoupled from coverage.
3. **Figure-3 baselines are absent from the released maze code.** The maze notebooks contain only LPM and pixel-curiosity; AMA/EDT/EME/Ensemble/IDF/RND are not present, so a full reproduction of the paper’s 7-method coverage ranking is not possible from this code. Our RND/ICM are our own ports.
4. **Coverage is a 2D projection.** Occupancy is binned on the floor plane; agent heading and the first-person view (what actually drives the pixel predictors) are not visualised here.